RAG Learning

Theme 1.1 How Markets Work — Edexcel B

1. Scarcity, Choice and Opportunity Cost

Scarcity: The fundamental economic problem — human wants are unlimited, but the resources (factors of production) available to satisfy those wants are finite. This forces societies, firms, and individuals to make choices about how to allocate resources.

1.1 The Economic Problem

Every society must answer three fundamental questions:

1. **What to produce?** — Which goods and services should be produced, and in what quantities?
2. **How to produce?** — What combination of resources and methods should be used?
3. **For whom to produce?** — How should the output be distributed among members of society?

1.2 Factors of Production

Factor	Description	Reward	Example
Land	All natural resources — minerals, water, climate, soil	Rent	Oil fields, farmland, rivers
Labour	The physical and mental effort of workers	Wages	Factory workers, teachers, programmers
Capital	Man-made aids to production (not money itself)	Interest	Machinery, factories, technology

© RAG Learning · Not for redistribution

Enterprise	The entrepreneur who organises the other factors and bears risk	Profit	Elon Musk, James Dyson

1.3 Opportunity Cost

Opportunity cost: The value of the next best alternative foregone when a choice is made. It is the real cost of any decision — measured not in money but in what you give up.

Opportunity cost applies at every level:

- **Individual:** A student choosing university gives up the salary they could have earned working full-time.
- **Firm:** A company investing in a new factory gives up the returns from investing that money elsewhere.
- **Government:** Spending on the NHS means less for education, defence, or infrastructure.

Case Study: HS2 Opportunity Cost

The UK government's HS2 high-speed rail project has cost estimates exceeding 100 billion pounds. Critics argue the opportunity cost is enormous — that money could fund hundreds of new hospitals, thousands of schools, or significant renewable energy investment. Supporters counter that HS2 will boost productivity and regional connectivity, generating returns that outweigh the opportunity cost. This illustrates how opportunity cost involves subjective judgement about the value of alternatives.

1.4 Free Goods vs Economic Goods

- **Free goods:** Have no opportunity cost — they are not scarce (e.g., air, sunlight).
- **Economic goods:** Have an opportunity cost because resources are used in their production (e.g., cars, healthcare).

2. Production Possibility Frontier (PPF)

PPF: A curve showing the maximum possible combinations of two goods that an economy can produce when all resources are used fully and efficiently, given current technology and resources.

2.1 Key Features of the PPF

- **Points on the curve** — productively efficient (all resources fully and efficiently employed).
- **Points inside the curve** — productively inefficient (unemployment or resource waste).
- **Points outside the curve** — currently unattainable with existing resources and technology.
- **Movement along the curve** — illustrates opportunity cost (more of one good means less of the other).
- The **concave (bowed-out) shape** reflects increasing opportunity cost — as you produce more of one good, resources become less well-suited and you sacrifice increasing amounts of the other.

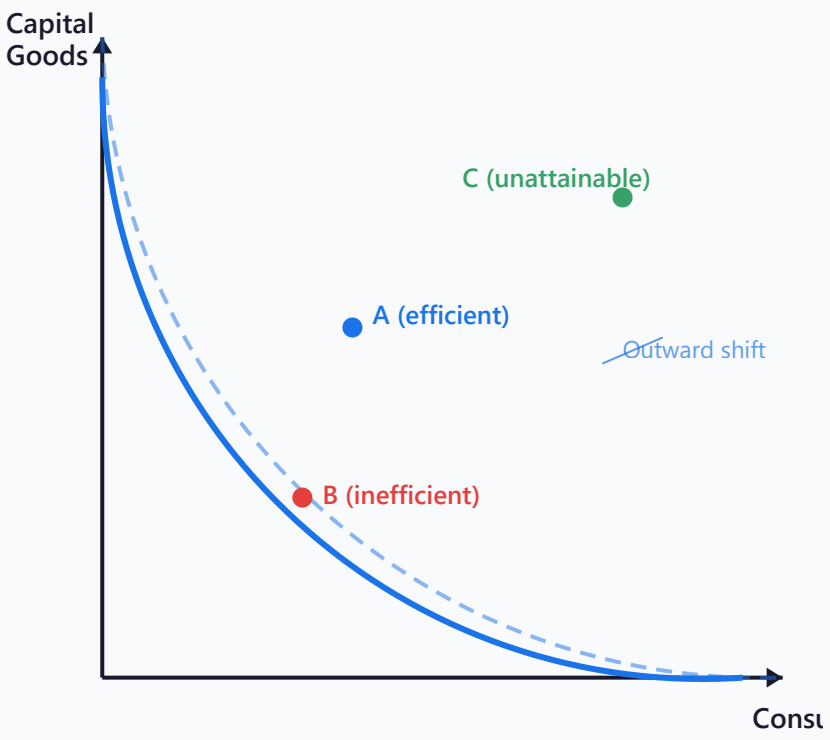

Figure 1: Production Possibility Frontier — quarter-circle shape showing increasing opportunity cost. A is efficient, B is inefficient, C is unattainable. Dashed line shows an outward shift from economic growth.

2.2 Shifts of the PPF

Outward Shift (Growth)	Inward Shift (Decline)
Increase in quantity or quality of resources	Natural disasters destroying resources
Technological improvements	Emigration of skilled labour
Investment in education and training	Depletion of natural resources
Discovery of new resources	War or conflict
Immigration of skilled workers	Pandemic reducing workforce

Exam Tip: A PPF shift can be asymmetric — if technology improves only in one industry, the PPF pivots outward on one axis only. Always label axes and mark key points when drawing PPF diagrams.

Case Study: COVID-19 and the UK PPF

The pandemic caused the UK economy to operate well inside its PPF in 2020 — GDP fell by 9.7%, the largest annual decline in over 300 years. Millions of workers were furloughed (idle resources) and many businesses closed temporarily. The economy moved from a point near the PPF to a point well inside it. Long COVID and early retirements may have also shifted the PPF slightly inward by reducing the effective labour supply.

3. Demand

Demand: The quantity of a good or service that consumers are willing and able to buy at a given price, in a given time period, *ceteris paribus* (all other things being equal).

3.1 The Law of Demand

There is an **inverse relationship** between price and quantity demanded — as price rises, quantity demanded falls, and vice versa. This creates a downward-sloping demand curve. Reasons:

- **Income effect:** When price rises, consumers' real income (purchasing power) falls, so they can afford less.
- **Substitution effect:** When price rises, consumers switch to cheaper alternatives.
- **Diminishing marginal utility:** Each additional unit consumed provides less extra satisfaction, so consumers only buy more at a lower price.

3.2 Individual vs Market Demand

Individual demand is one consumer's demand for a good. Market demand is the horizontal summation of all individual demand curves — at each price, we add together the quantities demanded by all consumers in the market.

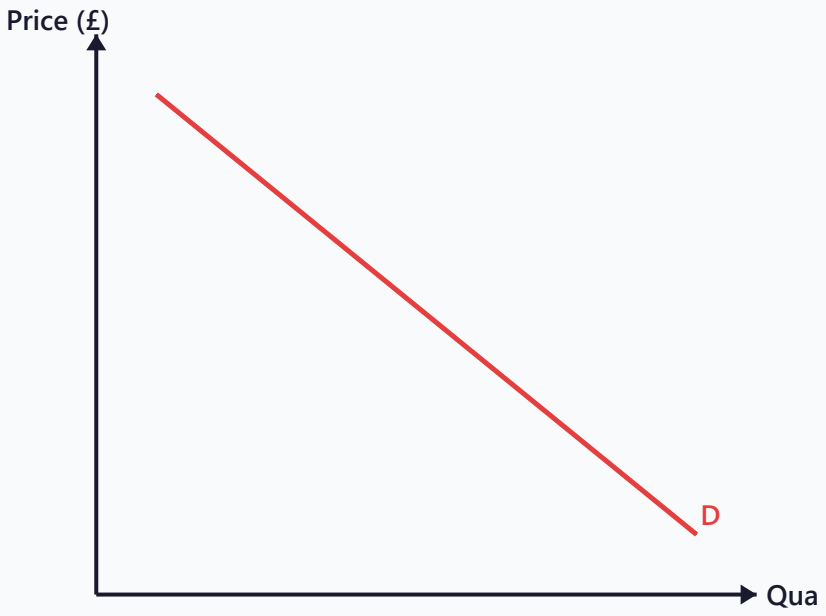

Figure 2: The demand curve — downward sloping, showing the inverse relationship between price and quantity demanded.

3.3 Movements Along vs Shifts of the Demand Curve

A **movement along** the demand curve (contraction or extension) is caused only by a change in the good's own price. A **shift** of the entire demand curve is caused by a change in a non-price factor.

Non-price Determinants of Demand (Shifters)

Factor	Rightward Shift (Increase in D)	Leftward Shift (Decrease in D)
Income	Rise in income (for normal goods)	Fall in income (normal goods); Rise in income (inferior goods)
Price of substitutes	Rise in price of a substitute	Fall in price of a substitute
Price of complements	Fall in price of a complement	Rise in price of a complement
Population	Increase in population	Decrease in population
Tastes / preferences	Fashions, advertising, trends increase desire	Negative publicity, health scares
Expectations	Expected future price rise boosts demand now	Expected future price fall reduces demand now
Interest rates	Lower interest rates (cheaper borrowing)	Higher interest rates

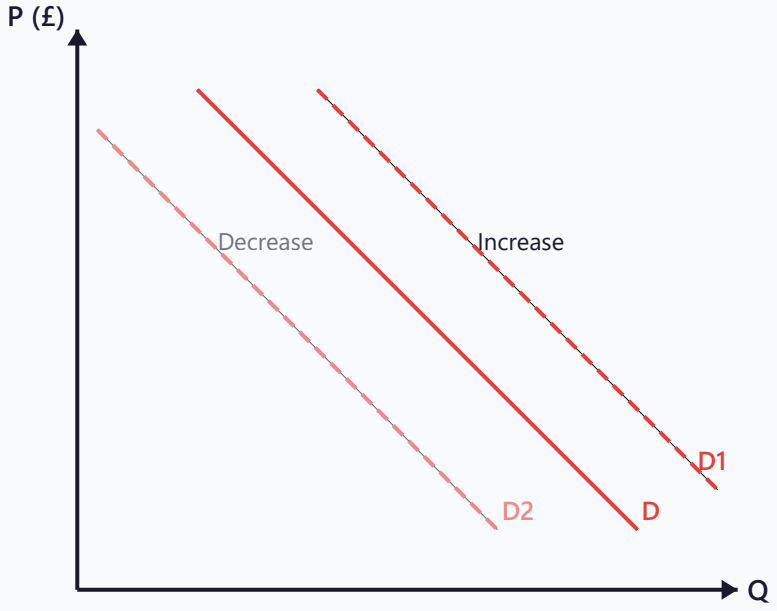

Figure 3: Shifts of the demand curve — D to D1 (increase), D to D2 (decrease).

Case Study: UK Electric Vehicle Demand

Demand for EVs in the UK surged from around 1,900 new registrations in 2013 to over 300,000 in 2023. Multiple demand shifters explain this: rising environmental awareness (tastes), government subsidies (lowering effective price), falling battery costs making EVs cheaper, expanding charging infrastructure (complementary goods), and the announced ban on new petrol/diesel car sales from 2035. This represents a significant rightward shift of the demand curve for EVs.

4. Supply

Supply: The quantity of a good or service that producers are willing and able to sell at a given price, in a given time period, *ceteris paribus*.

4.1 The Law of Supply

There is a **positive relationship** between price and quantity supplied — as price rises, quantity supplied rises. This creates an upward-sloping supply curve. Reasons:

- **Profit motive:** Higher prices mean higher potential profits, incentivising firms to produce more.
- **New entrants:** Higher prices attract new firms into the market.
- **Covering costs:** Higher prices allow higher-cost firms to enter the market profitably.

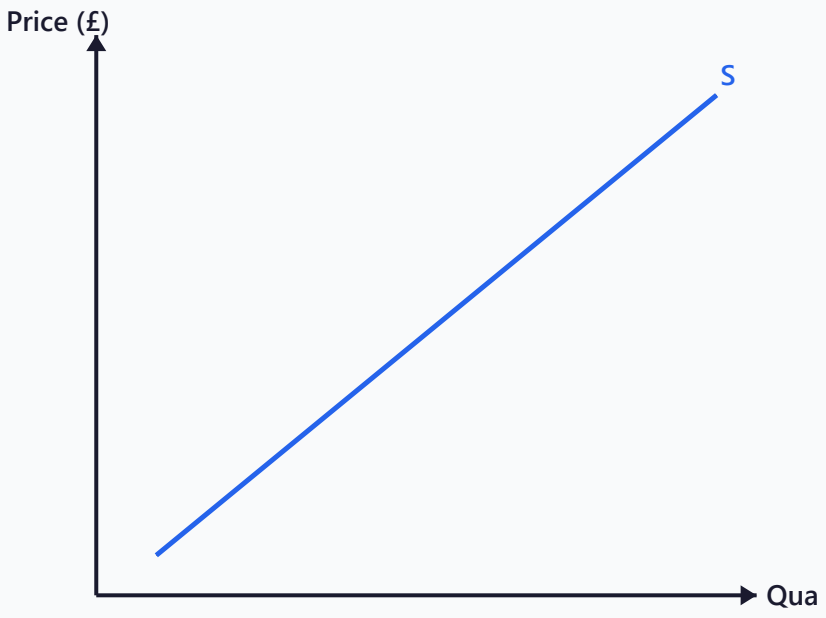

Figure 4: The supply curve — upward sloping, showing the positive relationship between price and quantity supplied.

4.2 Non-price Determinants of Supply (Shifters)

Factor	Rightward Shift (Increase in S)	Leftward Shift (Decrease in S)
Costs of production	Fall in wages, raw materials, rents	Rise in costs (e.g., oil price increase)
Technology	Improved technology reduces costs	Loss of technology (rare)
Indirect taxes	Removal/reduction of tax	Imposition/increase of indirect tax
Subsidies	Introduction/increase of subsidy	Removal/reduction of subsidy
Number of firms	New firms entering	Firms leaving
Weather/natural factors	Favourable weather (agriculture)	Drought, floods, natural disasters

Regulations	Deregulation	Increased regulation

5. Market Equilibrium

Market equilibrium: The point where quantity demanded equals quantity supplied at a given price. There is no tendency for price to change — the market clears with no excess demand or excess supply.

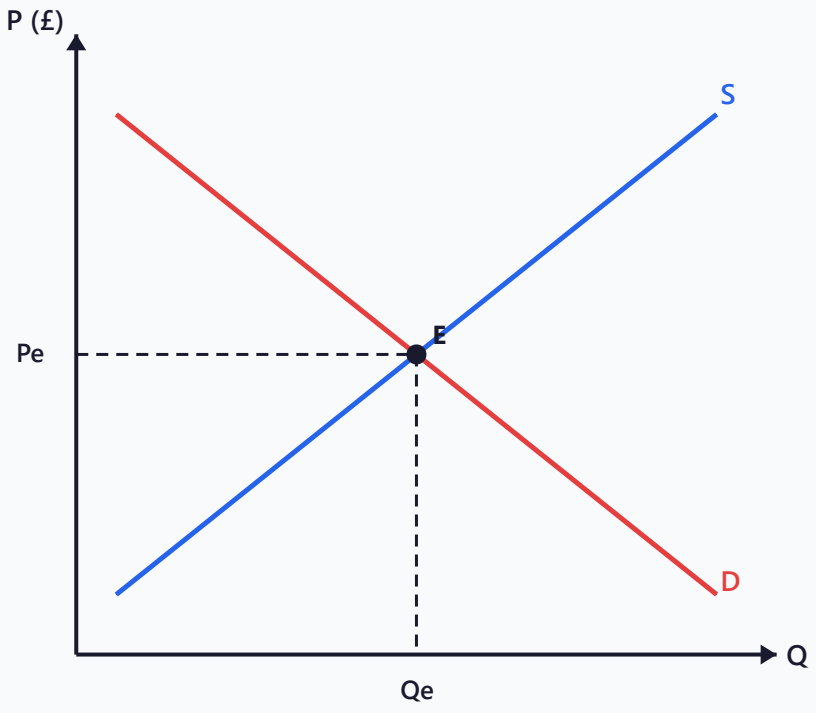

Figure 5: Market equilibrium where D and S intersect at price Pe and quantity Qe.

5.1 Disequilibrium

- **Excess demand (shortage):** Price below equilibrium, Qd > Qs. Consumers bid price up towards equilibrium.
- **Excess supply (surplus):** Price above equilibrium, Qs > Qd. Firms reduce price towards equilibrium.

5.2 Changes in Equilibrium

- **Increase in demand** (D shifts right) → higher equilibrium price and quantity.
- **Decrease in demand** (D shifts left) → lower equilibrium price and quantity.
- **Increase in supply** (S shifts right) → lower equilibrium price, higher quantity.
- **Decrease in supply** (S shifts left) → higher equilibrium price, lower quantity.

Exam Tip: When both demand and supply shift simultaneously, you can determine the direction of change in either price or quantity, but not both — one will be indeterminate. Always state this in your answer.

6. Consumer and Producer Surplus

Consumer surplus: The difference between what consumers are willing to pay and what they actually pay. Area below the demand curve and above market price.

Producer surplus: The difference between the price producers receive and the minimum they would accept. Area above the supply curve and below market price.

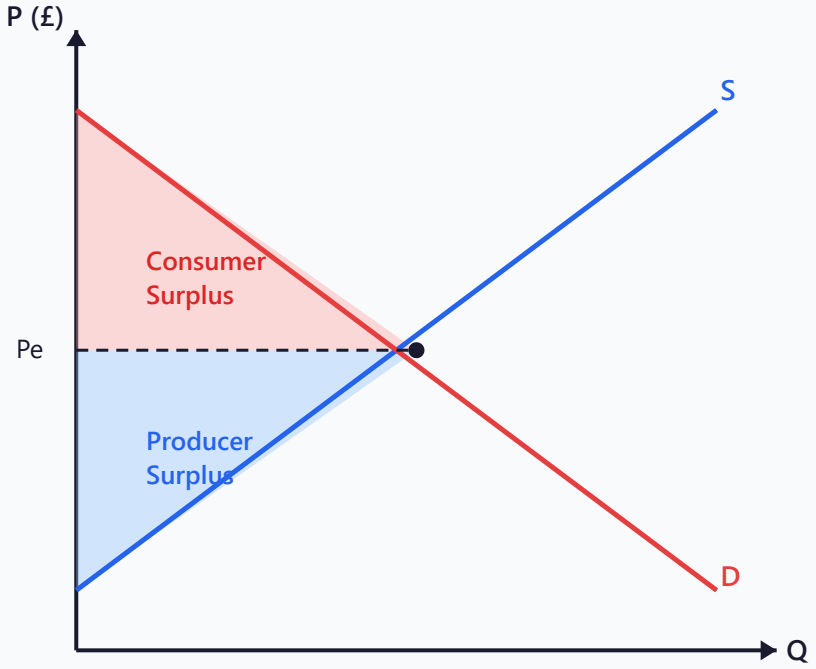

Figure 6: Consumer surplus (red area) and producer surplus (blue area) at equilibrium.

6.1 Total Economic Welfare

Total welfare = consumer surplus + producer surplus. At the free market equilibrium, total welfare is maximised — this is **allocative efficiency**. Any intervention moving price away from equilibrium creates a **deadweight welfare loss**.

6.2 Impact of Price Controls on Surplus

- **Price ceiling** (max price below equilibrium): increases consumer surplus for lucky buyers, but reduces producer surplus and creates deadweight loss and shortages.
- **Price floor** (min price above equilibrium): increases producer surplus for those who sell, but reduces consumer surplus and creates deadweight loss and surpluses.
- **Indirect tax:** Reduces both surpluses and creates deadweight loss.
- **Subsidy:** Increases both surpluses but at a cost to the taxpayer — may lead to over-production.

7. Elasticity

7.1 Price Elasticity of Demand (PED)

PED: A measure of the responsiveness of quantity demanded to a change in price. Always negative (inverse relationship) but we refer to the absolute value.

$$\text{PED} = \%\text{ change in quantity demanded} \div \%\text{ change in price}$$

Value	Term	Meaning	Demand Curve Shape
|PED| > 1	Elastic	Qd highly responsive — %ΔQd > %ΔP	Relatively flat
|PED| < 1	Inelastic	Qd not very responsive — %ΔQd < %ΔP	Relatively steep
|PED| = 1	Unit elastic	%ΔQd exactly equals %ΔP	Rectangular hyperbola
PED = 0	Perfectly inelastic	Qd does not change at all	Vertical line
PED = ∞	Perfectly elastic	Any price rise → Qd falls to zero	Horizontal line

Determinants of PED

- **Substitutes:** More substitutes = more elastic (consumers switch easily).
- **Necessity vs luxury:** Necessities are inelastic; luxuries are elastic.
- **Proportion of income:** Goods that take a large share of income are more elastic.
- **Time period:** Demand is more elastic in the long run (time to adjust).
- **Addictiveness:** Addictive goods (cigarettes, coffee) are inelastic.
- **Brand loyalty:** Strong loyalty makes demand more inelastic.
- **Market definition:** Broad markets (food) are more inelastic; narrow markets (organic avocados) more elastic.

PED and Total Revenue

- Demand **elastic** (|PED| > 1): price cut → TR rises (quantity gain outweighs price loss).
- Demand **inelastic** (|PED| < 1): price rise → TR rises (quantity loss is proportionally small).

© RAG Learning · Not for redistribution

- TR is maximised where |PED| = 1.

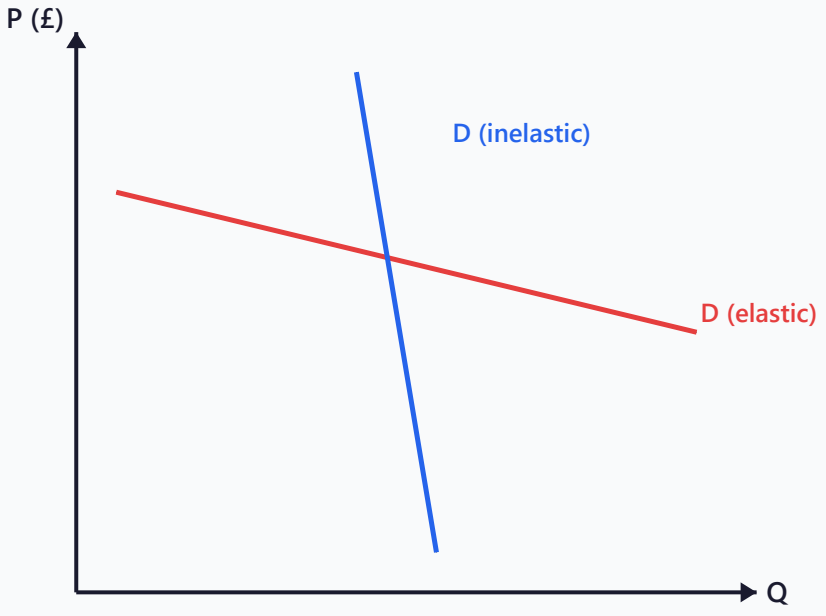

Figure 7: Elastic demand (relatively flat) vs inelastic demand (relatively steep).

Case Study: UK Sugar Tax and PED

In April 2018, the UK introduced the Soft Drinks Industry Levy — 18p/litre for drinks with 5-8g of sugar per 100ml, and 24p/litre above 8g. Research found that sugary drink purchases fell by about 30%. Many producers reformulated products to fall below the thresholds. The large response suggests demand for sugary drinks is somewhat elastic — consumers have substitutes (diet versions, water). However, some argue the main impact was supply-side (reformulation) rather than a demand response.

7.2 Income Elasticity of Demand (YED)

YED: A measure of the responsiveness of quantity demanded to a change in consumer income.

$$\text{YED} = \%\text{ change in quantity demanded} \div \%\text{ change in income}$$

Value	Type of Good	Example
YED > 1	**Luxury / superior good**	Designer clothes, overseas holidays, organic food
0 < YED < 1	**Necessity**	Bread, toothpaste, basic clothing
YED < 0	**Inferior good**	Budget brands, bus travel, instant noodles

Significance of YED for Businesses

- Luxury goods firms (YED > 1) are vulnerable to recessions — revenue falls sharply when incomes drop.
- Necessity firms (0 < YED < 1) are recession-proof — stable demand.
- Inferior goods firms (YED < 0) may benefit from recessions (e.g., Aldi and Lidl grew rapidly during 2008-09).
- Businesses use YED to forecast future demand based on expected economic growth.

7.3 Cross Elasticity of Demand (XED)

XED: A measure of the responsiveness of quantity demanded for good A to a change in price of good B.

$$\text{XED} = \%\text{ change in Qd of good A} \div \%\text{ change in price of good B}$$

Value	Relationship	Example
XED > 0	**Substitutes**	Butter and margarine
XED < 0	**Complements**	Cars and petrol
XED = 0	**Unrelated**	Pencils and milk

The larger the absolute value, the stronger the relationship. Firms use XED for pricing strategy, understanding competitive threats, and bundling decisions.

7.4 Price Elasticity of Supply (PES)

PES: A measure of the responsiveness of quantity supplied to a change in price. Always positive.

$$\text{PES} = \%\text{ change in quantity supplied} \div \%\text{ change in price}$$

Determinants of PES

- **Spare capacity:** Firms with unused capacity can increase output quickly — more elastic.
- **Availability of stocks:** Firms holding inventories respond quickly.
- **Ease of factor substitution:** If factors can be easily reallocated, supply is more elastic.
- **Time period:** Supply more elastic in the long run — firms can build new capacity.
- **Nature of product:** Agricultural goods with long growing seasons have inelastic supply; manufactured goods are more elastic.
- **Barriers to entry:** Low barriers allow new firms to enter quickly, making supply more elastic.

Case Study: UK Housing Supply Inelasticity

UK housing supply is notoriously price inelastic. Despite house prices rising over 60% between 2013 and 2023, housing completions increased only modestly — from around 120,000 to 180,000 per year. The barriers include strict planning regulations, limited land availability (especially in the South East), shortage of skilled construction workers, and long build times. This inelasticity means demand increases feed primarily into higher prices rather than more houses — a key driver of the UK housing affordability crisis.

8. The Price Mechanism

The price mechanism: The system by which changes in price act as signals to producers and consumers, provide incentives for economic agents to change behaviour, and ration scarce resources.

8.1 Three Functions of the Price Mechanism

Signalling

Prices convey information. A rising price signals high demand and encourages firms to allocate more resources to production. A falling price signals reduced demand and triggers reallocation of resources elsewhere. Prices coordinate the decentralised decisions of millions of agents.

Incentive

Prices create incentives for behavioural change. Higher prices incentivise producers to supply more (profit motive) and consumers to reduce consumption. Lower prices incentivise consumption and may push some firms out. This ensures resources flow to where they are most valued.

Rationing

The price mechanism rations scarce goods — those willing and able to pay receive the good; those who are not must go without. This allocates goods to those who value them most (in monetary terms), but raises equity concerns as it depends on ability to pay.

8.2 Adam Smith's "Invisible Hand"

Smith argued that individuals pursuing self-interest, guided by the price mechanism, produce an efficient resource allocation — as if by an "invisible hand." Without central planning, markets coordinate production and consumption to maximise welfare. However, this only holds when markets work well — market failures prevent efficient outcomes.

Evaluation: Limitations of the Price Mechanism

- Leads to inequality — rationing by price means the poor cannot afford essentials.
- Fails to account for externalities (e.g., pollution not reflected in prices).
- Underprovides public goods (non-excludable, non-rivalrous).
- Information failures mean prices may not reflect true costs and benefits.
- Monopoly power distorts prices and restricts output.
- Markets may adjust slowly — price signals take time to work through.

Exam Tip — Evaluation: Edexcel B examiners reward evaluation highly. Always consider: (1) short run vs long run effects, (2) the significance of the change, (3) who gains and who loses, (4) the

assumptions underlying your analysis, and (5) whether alternative policies might be more effective.